

Energy Bill Optimization

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Mode: Full Pipeline

1. Problem Formulation

Problem Type: Find
Domain: Integer Linear Programming

Universe

- Schedule: \{dishwasher, washer, dryer, ev_charger, pool_pump\}

Variables

Symbol	Meaning	Type / Range
t_i	start time for task i	\mathbb{R}_{\geq 0}

Structure

Schedule 5 household appliances (dishwasher, washer, dryer, EV charger, pool pump) across a 24-hour day to minimize daily electricity cost under time-of-use pricing. Each appliance runs once daily for a fixed duration at a fixed power draw. Some appliances (washer, dryer) cannot run overnight due to noise constraints.

Real-World Mapping

Real-World Concept	Mathematical Object
schedule	x: solution vector (schedule)

Constraints

- (1)

Time-of-use pricing

(2)

Binary assignment ILP

(3)

Noise constraints

(4)

Independent verification

(5)

Savings analysis
- electricity costs vary by hour; shifting loads

each appliance-start is a binary decision variable

washer and dryer restricted to daytime hours

6 checks (all scheduled, allowed hours, window)

optimal vs naive vs worst-case cost comparison

Objective

$$\min \sum_{i,j} c_{ij} x_{ij}$$

Approach

Suggested approach: ILP (PuLP/CBC, Branch & Bound)

Complexity class:

Available tools: ILP

2. Solution

Answer:	5-element schedule: dishwasher -> 3, washer -> 9, dryer -> 9, ev_charge
Objective Value:	4.695
Optimal:	Yes
Feasible:	Yes
Algorithm:	ILP (PuLP/CBC, Branch & Bound)
Time:	0.0051s
Certificate:	Branch-and-bound solved to optimality.

Solution Details

Schedule:

dishwasher -> 3
washer -> 9
dryer -> 9
ev_charger -> 2
pool_pump -> 1

Objective value: $z^* = 4.695$

Cost Breakdown:

dishwasher: 0.2880
washer: 0.0750
dryer: 1.5000
ev_charger: 2.3040
pool_pump: 0.5280

Comparison:

worst_case_cost: 14.8500
naive_cost: 11.8650
savings_vs_naive: 7.1700
savings_vs_worst: 10.1550

Method: Integer Linear Programming (ILP) via PuLP/CBC.

- Binary variables: $x[\text{appliance}, \text{start_hour}] = 1$ if appliance starts at that hour
- Objective: minimize sum over all appliance-hours of ($\text{power_kw} * \text{rate_at_hour}$)
- Constraints: each appliance starts exactly once, within its allowed hours, and

finishes within the 24-hour window

- Result: exact global optimum via Branch & Bound

Verification

✓ Feasibility

All constraints satisfied

✓ Optimality

ILP (PuLP/CBC, Branch & Bound)

3. Interpretation

The Question

Schedule 5 household appliances (dishwasher, washer, dryer, EV charger, pool pump) across a 24-hour day to minimize daily electricity cost under time-of-use pricing.

The Answer

The optimal solution has objective value $z^* = 4.695$.

What This Means

- Optimal start hour for each appliance
- Per-appliance cost breakdown
- 24-hour visual timeline showing appliance scheduling vs rate tiers
- Comparison against worst-case (all peak) and naive (typical daytime) schedules
- Monthly and annual projected savings
- Independent constraint verification (all PASS)

Integer Linear Programming (ILP) via PuLP/CBC.

- Binary variables: $x[\text{appliance}, \text{start_hour}] = 1$ if appliance starts at that hour
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- Result: exact global optimum via Branch & Bound

Recommendations

1. The solution is provably optimal; no further improvement is possible within this model.

Limitations

- Model assumes deterministic parameters; real-world variability not captured.

• Solution